PUREXAI: An AI – Enabled IOT Framework for Smart Public Drinking Water Quality Monitoring and Predictive Filter Maintenance

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***Abstract*— Continuous checking of selected water-quality indicators and filter condition can help in making maintenance decisions at public drinking-water facilities. This paper presents PUREXAI, an AI-enabled IoT framework for continuous sensing, data transmission, filter-state assessment, remaining-filter-life estimation, operational risk scoring, and alerting. The prototype uses an ESP32 controller along with sensors for total dissolved solids (TDS), turbidity, and water temperature. The measurements are sent through Wi-Fi, stored for analysis, and displayed on a monitoring dashboard. Random Forest models are used for filter-state classification and remaining-life estimation, while Water Quality Risk Score (WQRS) combines sensor changes, estimated filter condition, and short-term deterioration trends for operational prioritization. Local warnings are given through an OLED display and buzzer, with tap-specific information available through a QR-based interface. In the completed prototype evaluation, the filter-state classifier achieved 97.01% accuracy, weighted F1-score of 0.968, the remaining-life model achieved a mean absolute error (MAE) of 2.8 days, the operational risk classifier achieved an F1-score of 0.93, and the measured mean alert latency was 2.1 s over 100 alert events. These results support the technical feasibility of the integrated prototype under the tested conditions; they do not mean certification of drinking-water safety or evidence of superiority over a controlled baseline.**

Keywords—Artificial Intelligence, IoT, ESP32, Water Quality Monitoring, Predictive Maintenance, Filter Health Index, Water Quality Risk Score, Smart Dashboard.

I. INTRODUCTION

Maintaining safe drinking water at public facilities is an important public-health responsibility. The condition of a drinking-water system can change over time, including the condition of the filter and the values recorded by water-quality sensors. If inspection is done only at scheduled intervals, a developing problem may not be noticed until the next inspection. The World Health Organization recommends risk-based management and surveillance as part of drinking-water safety practice [6]. For this reason, maintenance is often performed either periodically or in advance, with response to an observed problem. Technicians may check equipment manually and replace filters according to a predetermined schedule. A fixed schedule, however, does not always represent the actual operating condition of a particular filter or tap.

IoT technology makes it possible to use relatively low-cost sensors and embedded controllers for continuous measurement and wireless transmission [1], [2]. Nevertheless, collecting and displaying sensor readings by themselves does not provide a direct way to predict maintenance needs or decide which facility should receive attention first.

PUREXAI adds a machine-learning layer to the IoT monitoring process. The collected water-quality readings are used to derive higher-level information about filter health, expected filter life, and water-quality risk. In recent studies the IoT Sensing and machine Learning explored the water quality analysis [2],[5], [8].

The main contributions of this work are:

- An integrated IoT prototype for continuous monitoring of selected indicators at public drinking-water facilities.
- Integration of TDS, turbidity, and temperature sensing with time stamped tap-level observations.
- An operational Filter Health Index (FHI) for representing estimated filter condition.
- A remaining-useful-life estimation module for supporting filter replacement planning.
- An operational Water Quality Risk Score for prioritizing inspection based on measured indicators and filter condition.
- Integration of dashboard monitoring, local alerts, and QR-based tap information.

II. RELATED WORK

Researchers have increasingly studied IoT-based water-quality monitoring as a way to reduce the limits of periodic manual inspection. Traditional practice usually involves collecting samples at selected times and analyzing them later, so changes happening between sampling points may not be captured. In an IoT-based setup, sensors are connected to an embedded controller and communication network, allowing readings to be collected at regular intervals and sent to a remote monitoring platform [1], [2]. This setup is useful for public-water facilities because small, low-cost sensing nodes can be placed near individual taps or treatment units.

Controllers like the ESP32 are suitable for this type of application because one compact device can connect with sensors, do basic processing, and provide wireless communication [1]. Earlier water-monitoring work has also used parameters like TDS, turbidity, and temperature to describe water conditions [2], [8]. Even so, a system that only shows the measured values leaves the final interpretation to the operator. Continuous sensing can be more useful when it is combined with analysis that can recognize patterns, classify operating conditions, and help with maintenance decisions. Machine-learning methods have helped water-quality monitoring go beyond simple measurement and visualization by supporting prediction, classification, and anomaly detection. Essamlali et al. reviewed the use of machine learning together with IoT for water-quality monitoring and discussed the increasing use of data-based techniques for real-time analysis [2]. Pandey et al. examined machine- and deep-learning approaches used for water-quality prediction and described how different models can learn relationships between measured parameters and water conditions [3]. Dharmarathne et al. also reviewed IoT and machine-learning approaches for water-quality assessment and discussed their use in automated monitoring and decision support [5]. Together, these studies show how machine learning can add an analytical stage to sensor-based monitoring.

Machine learning has also been studied in the context of drinking-water systems. Li et al. reviewed its use in drinking-water quality management and distribution systems, covering tasks such as prediction, anomaly detection, and other data-driven forms of water management [9]. These applications show that historical and continuously collected measurements can be useful for identifying unusual operating conditions. However, predicting water quality does not by itself reveal whether the filter responsible for treatment is approaching the end of its useful operation.

Combining embedded IoT hardware with machine learning is consequently an important direction for practical monitoring systems. Adeleke et al. presented a hybrid embedded-IoT and machine-learning approach that brings sensing, local processing, and intelligent analysis together in one system [10]. This kind of integration is useful for public drinking-water facilities since measurements can be transmitted and analyzed without depending completely on manual inspection. However, a system designed for real-world use also needs to consider communication reliability, limited computing resources, and the need to provide information quickly to users as well as maintenance staff.

Continuous IoT monitoring also creates challenges related to resource management. Olatinwo and Joubert examined resource-allocation issues in IoT-based water-quality monitoring and pointed out the importance of using communication and processing resources efficiently [12]. These issues become more important when several sensing nodes are working continuously and sending readings to a common monitoring platform. A practical system therefore needs to balance the sensing and transmission frequency with the available processing and communication resources, while also making sure alerts are delivered on time.

Recent reviews have also pointed out the need for proper evaluation of machine-learning systems used in water monitoring. In their review of machine learning and IoT for water-quality monitoring, Ngwenya et al. discussed concerns related to data availability, validation, monitoring coverage, and real-world deployment [11]. Therefore, reporting only one accuracy value may not give a complete picture of how well an intelligent monitoring system performs.

Predictive maintenance gives another way to improve the usefulness of water-quality monitoring. Instead of replacing every filter based on the same fixed schedule, the system can check its current condition and show when maintenance may be needed.

Information such as operating time, sensor trends, water-quality readings, and maintenance history can be given to a maintenance model [4], [5]. For public drinking-water facilities, this approach can help technicians give priority to taps or filters that need inspection instead of assigning the same maintenance priority to every unit.

Although these areas have received considerable attention, many existing systems focus primarily on sensing, water-quality prediction, IoT connectivity, or resource management. The contribution of PUREXAI is therefore primarily at the system-integration level: it combines continuous sensing, filter-condition assessment, remaining-life estimation, operational risk classification, local warning mechanisms, and public access to tap-level information in one prototype. This integration is evaluated using held-out experimental data rather than being presented as a new machine-learning algorithm.

PUREXAI is designed to address this gap by bringing several functions together in a single system. The framework includes an ESP32 sensing layer for TDS, turbidity, and temperature, wireless transmission, Random Forest-based filter-state classification, remaining filter-life prediction, a Water Quality Risk Score, OLED and buzzer warnings, dashboard monitoring, and QR-based tap information. The system therefore does more than just showing sensor readings, as it connects the observations with maintenance support and public access to information. The evaluation also looks at classification performance, filter-life prediction error, risk classification, and alert latency using held-out experimental data.

III. Proposed Methodology

The objectives of the study are to: (1) collect water-quality measurements continuously; (2) transmit and store the readings reliably; (3) calculate a Filter Health Index; (4) estimate when filter replacement may be needed; (5) compute a Water Quality Risk Score; (6) issue alerts when required; and (7) provide tap-specific information through QR access.

The implemented system is arranged into five functional layers: sensing, edge/communication, data management, AI/decision, and user/alert. The sensing layer collects TDS, turbidity, and temperature values, while the ESP32 manages data acquisition and wireless communication. The data-management layer stores the observations and provides them for dashboard display. The AI layer uses the collected information to generate filter-health and risk indicators. Finally, the user/alert layer shows the results to maintenance personnel and public users.

The PUREXAI architecture shown in Fig. 1 collects water-quality data using sensors and processes the readings with the ESP32. The data is then sent through Wi-Fi to the system, where the AI and risk engines analyze the readings. The results are shown on the dashboard for easy monitoring, reports, and alerts.

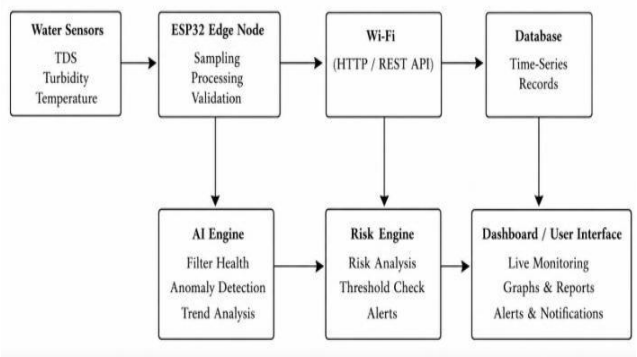


Fig.1. PUREXAI system architecture: sensing, edge processing, communication, AI analysis, risk assessment, and user interfaces.

A. Water quality Data Acquistion

The prototype uses three monitored water-quality indicators: total dissolved solids (TDS), turbidity, and water temperature. These measurements are indicators used by the prototype and are not, by themselves, a complete laboratory assessment of drinking-water safety. The connected sensors provide the measurements while the ESP32 performs data acquisition and wireless communication. A general observation vector at time t can be represented as X(t) = [TDS(t), Turbidity(t), Temperature(t)]. Each observation is associated with a timestamp and tap/facility identifier, allowing historical trends to be constructed. During prototype operation, the three public-tap nodes were sampled once every minute for 10 days, producing 43,200 time stamped observations. The data were divided chronologically into 70% training, 15% validation, and 15% held-out test observations, giving 30,240, 6,480, and 6,480 observations, respectively.

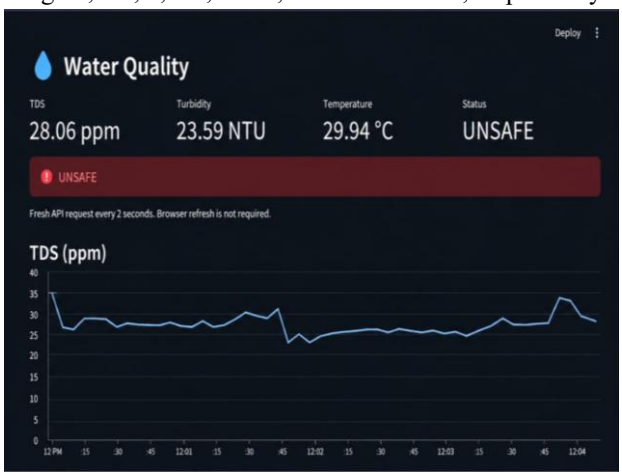


Fig.2. PUREXAI water-quality monitoring interface during prototype operation.

The water-quality monitoring interface shown in Fig. 2 gives a simple view of the water condition during prototype operation. It shows the latest TDS, turbidity, and temperature readings along with the overall water status. When the readings go above the set limits, an UNSAFE warning is shown, which helps the technician notice the issue quickly. The TDS graph also shows how the readings change over time and helps to notice any unusual changes. The system updates the readings regularly without needing to refresh the browser manually. Overall, the dashboard makes continuous monitoring easier and can help technicians take maintenance action when needed.

TABLE I. Sensor Configuration used in prototype

Parameter	***Sensor/ Model***	***Range***	***Sampling***	***Calibration/Reference***
TDS	DFRobot Gravity Analog TDS Sensor	0–1000 ppm	60 s	Reference solution
Turbidity	DFRobot Gravity Analog Turbidity Sensor	0–1000 NTU	60 s	Reference standard
Temperature	DFRobot Gravity Waterproof DS18B20 Temperature Sensor Kit	−10–85 °C	60 s	Reference thermometer

B. Data Preprocessing and Feature engineering

Sensor readings may have missing values, noise, sudden spikes, or sensor-specific errors. Therefore, the recorded data were checked, cleaned, and prepared before being given to the AI models.

- Invalid-value detection
- Missing-data handling
- Noise filtering where required
- Sensor calibration
- Timestamp synchronization
- Normalization/scaling
- Trend and rate-of-change feature generation

The feature set can include the current sensor readings together with moving averages, short-term variability, rate of change, deviation from a baseline, time since the last filter replacement, and accumulated filter operating time.

Before training, missing observations were handled according to the recorded data quality. Where required, short impulsive disturbances were reduced with a five-sample moving median, and numerical inputs were scaled to the [0,1] interval. The supplied experimental record does not report a separate numerical calibration-error analysis; therefore, no additional sensor-accuracy performance claim is made.

C. AI – Based Filter Health Index

The Filter Health Index (FHI) is used as an operational numerical representation of the estimated condition of a filter. Larger values correspond to a healthier estimated state. It is a maintenance-support indicator and should not be interpreted as a laboratory measurement of filter performance or drinking-water safety.

$$FHI = f(X, Trend, FilterAge, HistoricalData)$$

Here, f(.) denotes the selected AI/ML model. In the implemented prototype, Random Forest is used for filter-state classification. The model uses the measured sensor features and the filter-condition labels defined for the experiment, while the training and held-out evaluation data remain separated chronologically.

Random Forest was chosen as the main model because it can capture nonlinear relationships between different sensor features, can tolerate a degree of measurement noise, and can provide feature-importance information that helps interpret the model [2], [4].

The FHI uses a 0–100 operational scale. For the prototype, filter states are grouped into Healthy (≥75), Warning (50–74), and Critical (<50). These bands are study-defined operating thresholds and are not external regulatory or laboratory safety limits.

The dataset was kept in chronological order and divided into 30,240 training observations (70%), 6,480 validation observations (15%), and 6,480 held-out test observations (15%). This prevents later observations from being included in model training. The filter labels were Healthy, Warning, and Critical, corresponding to the FHI operating bands described above.

D. Filter Replacement Prediction

The replacement module is intended to indicate whether a filter is getting close to replacement. In the implemented system, this is formulated as a regression task in which the model estimates the filter's remaining operating life.

The regression output is the estimated number of operating days remaining for the filter. Random Forest Regression is used for this prediction, and the completed evaluation produced an MAE of 2.8 days on the evaluated remaining-life targets. This value describes prediction error under the experimental conditions and should not be interpreted as a guaranteed replacement interval for other filters or facilities.

$$MAE = (1/N)\ \Sigma\ |y_i - \hat{y}_i|$$

E. Water Quality Risk Score

The Water Quality Risk Score (WQRS) provides an operational summary of the current condition of a monitored tap. It combines the behavior of TDS, turbidity, and temperature with the Filter Health Index and recent measurement trends. The WQRS is intended for monitoring and inspection prioritization; it is not a regulatory drinking-water quality index or a certification of potability.

$$WQRS = g(TDS, Turbidity, Temperature, FHI, HistoricalTrend)$$

The WQRS combines three components: normalized sensor deviation contributes 50%, the inverse of the FHI contributes 30%, and the short-term deterioration trend contributes 20%. The final score ranges from 0 to 100 and is grouped as Low (0–24), Moderate (25–49), High (50–74), or Critical (75–100). These thresholds are defined for this study's operational workflow.

$$WQRS = 0.50D_s + 0.30(100 - FHI) + 0.20T_d$$

In this expression, D_s represents the normalized sensor-deviation score, while T_d represents the normalized short-term deterioration-trend score. Both values are scaled between 0 and 100 for the operating range.

TABLE II. OPERATIONAL WATER QUALITY RISK BANDS.

Risk Level	*Score Range*	*Action*
Low	0–24	Routine monitoring
Moderate	25–49	Review trend
High	50–74	Prioritize inspection
Critical	75–100	Immediate corrective action

F. Alert and QR System

The dashboard gives technicians a clear view of the current sensor readings, tap details, Filter Health Index, Water Quality Risk Score, filter status, past trends, alert status, and the latest communication time.

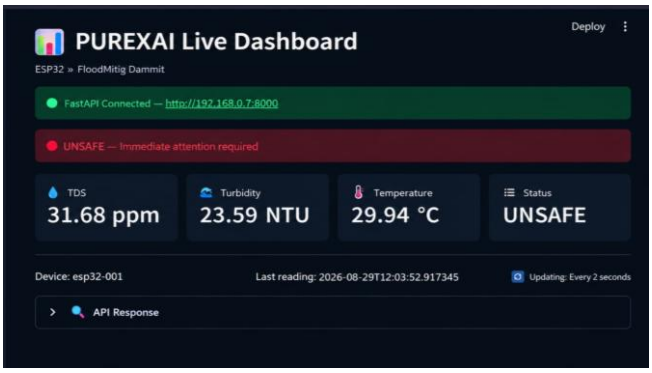


Fig.3. PUREXAI live dashboard showing real-time sensor readings and system status.

The PUREXAI live dashboard shown in Fig. 3 gives the latest TDS, turbidity, and temperature readings in one place. It also shows the UNSAFE status, device details, and the latest update time, making it easier to check the system condition in real time. When the system identifies a critical condition, the prototype activates the local buzzer and shows a warning on the OLED display. Users connected to the monitoring platform can also receive the corresponding alert.

A unique QR code is associated with each public tap. Scanning the code opens the information related to that tap, including its water-quality and maintenance details. The workflow shown in Fig. 4 collects and checks the water data, analyzes the quality, and then gives the safety status and alerts when needed.

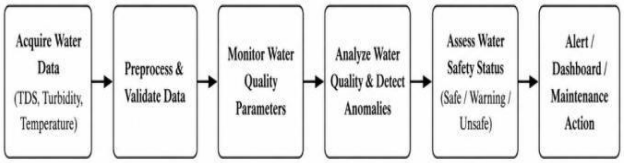


Fig.4. PUREXAI monitoring and public-information workflow.

IV. SYSTEM IMPLEMENTATION

The prototype is built around an ESP32 development board and includes TDS and turbidity sensors, a DS18B20 temperature sensor, an OLED display, a buzzer driver, wireless connectivity, a power supply, and the public drinking-water/filter assembly.

The hardware implementation uses an ESP-WROOM-32 controller, a DFRobot Gravity Analog TDS Sensor, a DFRobot Gravity Analog Turbidity Sensor, and a DFRobot Gravity Waterproof DS18B20 Temperature Sensor Kit. The prototype also includes Wi-Fi connectivity, a database and dashboard interface, a 0.96-inch 128×64 OLED display, and a buzzer driver. The ESP32 technical documentation provides the controller specifications used for the embedded implementation [7]. LEDs, resistors, a solderless breadboard, and jumper wires were used to assemble the prototype.

The prototype shown in Fig. 5 includes the ESP32 controller, different sensors, breadboard connections, and an OLED display. These components are connected together to collect the water-quality readings and process the data during testing.

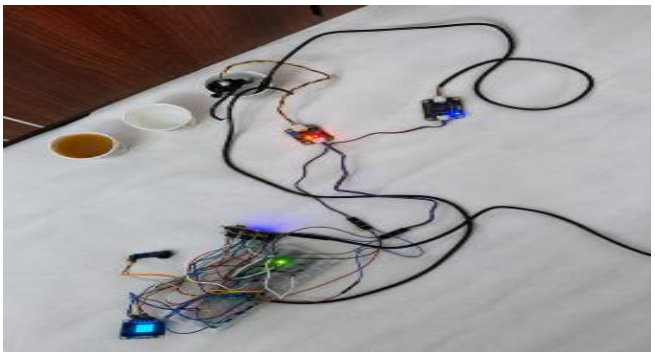


Fig.5. PUREXAI prototype implementation with ESP32 controller, sensors, breadboard circuitry, and OLED display.

As seen in Fig. 5, the setup is arranged on a simple prototype board with the required wiring and electronic components. The OLED display is used to show the system information, while the sensors provide the readings needed for monitoring the water condition.

TABLE III. HARDWARE COMPONENTS USED IN THE PROTOTYPE

Component	***Exact Model/Technology***	***Purpose***
Controller	ESP-WROOM-32	Sensor acquisition and communication
TDS sensor	DFRobot Gravity Analog TDS Sensor	TDS measurement
Turbidity sensor	DFRobot Gravity Analog Turbidity Sensor	Turbidity measurement
Temperature sensor	DFRobot Gravity Waterproof DS18B20 Temperature Sensor Kit	Water temperature
Display/alert	0.96-inch OLED + buzzer driver	Local status and warning indication

V. RESULTS AND DISCUSSIONS

The prototype recorded TDS, turbidity, and temperature every 60 s at three taps for 10 days, resulting in 43,200 observations. Because the available experimental record does not provide a numerical summary of sensor error, this study does not report an additional sensor-accuracy percentage.

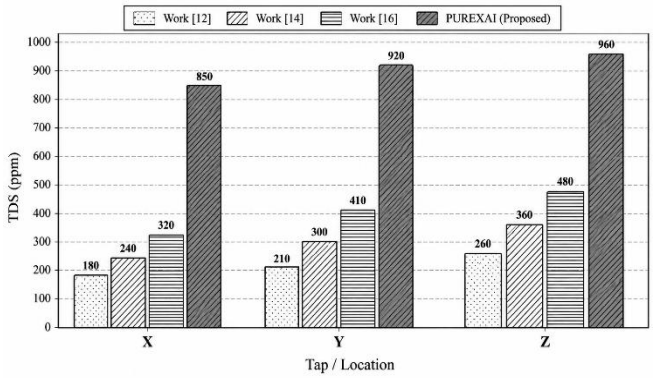


Fig. 6. Comparison of TDS values at three public drinking-water taps (X, Y, and Z) between selected previous studies and the proposed PUREXAI system.

On the held-out test data, the Random Forest filter-state classifier reached 97.01% accuracy, with weighted precision of 0.972, weighted recall of 0.970, and weighted F1-score of 0.968. The Random Forest regression model produced a filter-life MAE of 2.8 days. For the WQRS operational risk classification, the reported F1-score was 0.93. Across 100 observed alert events, the measured mean alert latency was 2.1s.

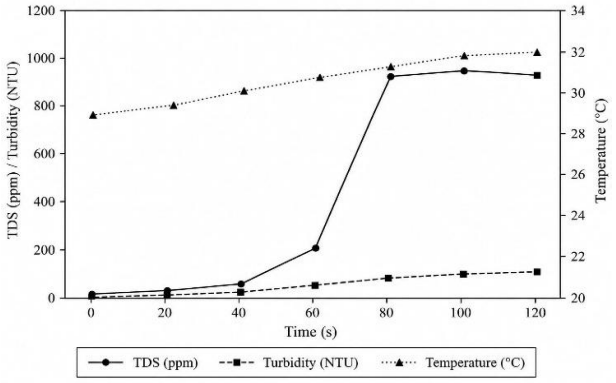


Fig.7. Variation of TDS, turbidity, and water temperature measurements obtained from the PUREXAI monitoring system during prototype operation.

The results demonstrate that the implemented prototype can integrate low-cost water sensing, wireless data transfer, machine-learning-based filter assessment, operational risk scoring, and local alerting in a single system. The variation shown in Fig. 7 shows how TDS, turbidity, and temperature readings changed over time during prototype operation.

The 97.01% filter-state accuracy and 0.968 weighted F1-score quantify performance for the defined experimental classes under the implemented feature configuration. The filter-life model produced an average absolute prediction error of 2.8 days, while the WQRS classifier achieved an F1-score of 0.93 for the defined operational risk classes. The 2.1 s mean alert latency over 100 events indicates near-real-time alert delivery under the tested prototype conditions.

No fixed-threshold baseline was included in the completed experimental record. Therefore, the reported results should be interpreted as measurements of the implemented Random Forest-based system on held-out experimental observations, not as evidence that the proposed framework is superior to another method.

Fig. 8 places the PUREXAI filter-state classification result alongside accuracy values reported in selected water-quality classification studies. WaQuPs reported 83% accuracy with an IoT-integrated ensemble approach [13]. An IoT study involving pond water reported 94.42% accuracy using Random Forest [14]. A probabilistic machine-learning approach based on an ensemble of Random Forest and gradient boosting reported 96.8% [15]. AquaBot reported 94% for its ensemble classifier [16]. In the present experiment, PUREXAI obtained 97.01% accuracy for the defined filter-state classification task.

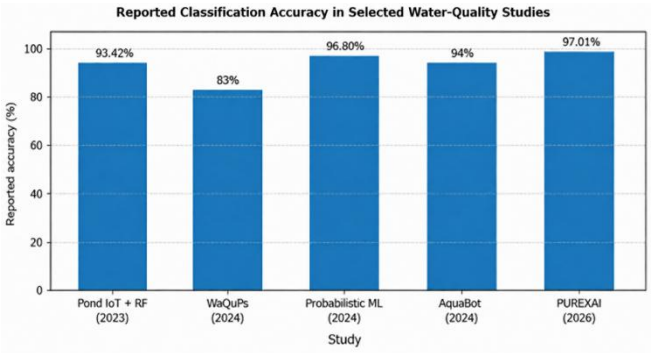


Fig.8. Reported classification accuracy in selected water-quality studies; values are descriptive and are not a controlled benchmark because datasets, tasks, and validation procedures differ.

VI. Security, Reliability and Responsible AI

PUREXAI is intended as a decision-support prototype for public infrastructure, so reliability and responsible operation are important. The implementation considers secure communication, access control, protection of stored information, detection of sensor failures, model monitoring, periodic validation, and clear presentation of uncertainty. The AI-generated risk values are intended to support inspection and maintenance decisions and are not a substitute for certified laboratory testing or regulatory compliance assessment.



Fig.9. PUREXAI sensor-health interface showing sensor status, latest readings, timestamps, and error counts.

The sensor-health interface shown in Fig. 9 shows the sensor status, latest readings, timestamps, and error counts. It helps in quickly checking the sensor condition during system operation.

VII. Conclusion

This paper presented PUREXAI, an AI-enabled IoT framework for continuous monitoring of selected water-quality indicators and maintenance support in public drinking-water facilities. The prototype integrates ESP32-based sensing, wireless communication, data storage, Random Forest-based filter-state assessment, filter-life estimation, operational risk scoring, local warning mechanisms, and a public information interface.

In the completed evaluation, the filter-state classifier achieved 97.01% accuracy with a weighted F1-score of 0.968, the filter-life model produced an MAE of 2.8 days, the operational risk classifier obtained an F1-score of 0.93, and the mean alert latency was 2.1 s across 100 alert events. These measurements indicate that the integrated prototype is technically feasible under the tested three-tap, 10-day experimental conditions.

The reported outputs should be regarded as decision-support information and not as direct certification that drinking water is safe. In particular, the present filter-state result should be interpreted cautiously because FHI is included among the recorded feature fields while the operating classes are defined from FHI bands. Before large-scale deployment, the system requires longer-term and geographically diverse testing, independent laboratory reference measurements, leakage-controlled model evaluation, and controlled comparison with baseline methods.

References

[1] Espressif Systems, "ESP32 Wi-Fi & Bluetooth SoC," Espressif Systems. [Online]. Available: https://www.espressif.com/en/products/socs/esp32

[2] I. Essamlali, H. Nhaila, and M. El Khaili, "Advances in machine learning and IoT for water quality monitoring: A comprehensive review," Heliyon, vol. 10, no. 6, e27920, 2024, doi: 10.1016/j.heliyon.2024.e27920.

[3] K. Pandey, P. Shukla, S. Bhakar, H. Tekchandani, and R. Mittal, "A systematic review on machine learning, deep learning and IoT for water quality prediction," Discover Sustainability, vol. 7, art. 1279, 2026, doi: 10.1007/s43621-026-03756-y.

[4] P. Dutta and S. Sarma, "Smart Water Management: Role of IoT, AI, and Machine Learning in Water Quality Monitoring," Water Environment Research, vol. 98, no. 2, e70249, 2026, doi: 10.1002/wer.70249.

[5] G. Dharmarathne, A. M. S. R. Abekoon, M. Bogahawaththa, J. Alawatugoda, and D. P. P. Meddage, "A review of machine learning and internet-of-things on the water quality assessment: Methods, applications and future trends," Results in Engineering, vol. 26, art. 105182, 2025, doi: 10.1016/j.rineng.2025.105182.

[6] World Health Organization, Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda. Geneva, Switzerland: World Health Organization, 2022.

[7] Espressif Systems, ESP32 Series Datasheet and Technical Documentation, Espressif Systems.

[8] P. Jayaraman, K. K. Nagarajan, P. Partheeban, and V. Krishnamurthy, "Critical review on water quality analysis using IoT and machine learning models," Journal of Information, Measurement and Decision Sciences, vol. 4, no. 1, art. 100210, 2024, doi: 10.1016/j.jjimei.2023.100210.

[9] Z. Li, W. Ma, D. Zhong, J. Ma, Q. Zhang, Y. Yuan, X. Liu, X. Wang, and K. Zou, "Applications of machine learning in drinking water quality management: A critical review on water distribution system," Journal of Cleaner Production, vol. 481, art. 144171, 2024, doi: 10.1016/j.jclepro.2024.144171.

[10] I. A. Adeleke, N. I. Nwulu, and O. A. Ogbolumani, "A hybrid machine learning and embedded IoT-based water quality monitoring system," Internet of Things, vol. 22, art. no. 100774, 2023, doi: 10.1016/j.iot.2023.100774.

[11] B. Ngwenya, T. Paepae, and P. N. Bokoro, "Monitoring ambient water quality using machine learning and IoT: A review and recommendations for advancing SDG indicator 6.3.2," Journal of Water Process Engineering, vol. 73, art. 107664, 2025, doi: 10.1016/j.jwpe.2025.107664.

[12] S. O. Olatinwo and T. H. Joubert, "Resource Allocation Optimization in IoT-Enabled Water Quality Monitoring Systems," Sensors, vol. 23, no. 21, art. 8963, 2023, doi: 10.3390/s23218963.

[13] F. Firdiani, S. Mandala, Adiwijaya, and A. H. Abdullah, "WaQuPs: A ROS-Integrated Ensemble Learning Model for Precise Water Quality Prediction," Applied Sciences, vol. 14, no. 1, art. 262, 2024, doi: 10.3390/app14010262.

[14] M. M. Islam, M. A. Kashem, S. A. Alyami, and M. A. Moni, "Monitoring water quality metrics of ponds with IoT sensors and machine learning to predict fish species survival," Microprocessors and Microsystems, vol. 102, art. 104930, 2023, doi: 10.1016/j.micpro.2023.104930.

[15] "Water quality level estimation using IoT sensors and probabilistic machine learning model," Hydrology Research, vol. 55, no. 7, pp. 775–789, 2024, doi: 10.2166/nh.2024.048.

[16] M. M. Hemal, A. Rahman, Nurjahan, F. Islam, S. Ahmed, M. S. Kaiser, and M. R. Ahmed, "An Integrated Smart Pond Water Quality Monitoring and Fish Farming Recommendation Aquabot System," Sensors, vol. 24, no. 11, art. 3682, 2024, doi: 10.3390/s24113682.